Face Recognition & Mask Detection System

Automated Identity Verification Using Computer Vision

Real-time Face Recognition
The system identifies individuals instantly using advanced computer vision algorithms capable of operating at 30 FPS.

Mask Detection Capability
Integrated mask detection ensures compliance with health protocols using custom-trained Haar Cascade classifiers.

Multiple Algorithm Implementation
Combines EigenFace, FisherFace, and LBPH methods for robust and accurate recognition across varied conditions.

Contactless Verification
Provides secure, touch-free identification, ideal for high-traffic environments such as offices, campuses, and security zones.

Why Face Recognition & Mask Detection?

Introduction & Problem Statement

- **Need for Contactless Verification:** Traditional identity verification systems involve physical contact, posing hygiene and efficiency concerns in post-pandemic environments.
- **Automation and Health Safety:** Demand for touch-free solutions drives innovation in face recognition technologies ensuring minimal human intervention.
- **Technical Challenges:** The system must perform real-time processing under variable lighting, angles, and partial occlusions caused by masks.
- **Accuracy and Reliability:** Achieving high accuracy while maintaining fast response time is crucial for large-scale deployment in institutions and workplaces.

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System Overview

Automated Face Recognition and Mask Detection

Purpose
Develop an automated system that performs real-time face recognition and mask detection for contactless identity verification.

Key Features
Incorporates multiple algorithms, supports dual cameras for entry and exit points, and maintains automated data logging.

Real-Time Operation
Processes video feeds at approximately 30 frames per second while maintaining high recognition accuracy.

Mask Compliance Monitoring
Provides continuous supervision of mask usage with instant alerts or logs for non-compliance events.

System Components

Modules of the Face Recognition & Mask Detection System

Image Collection Module
Captures and preprocesses face images, performing automatic face detection and normalization to ensure uniformity across samples.

Model Training Module
Trains multiple algorithms including EigenFace, FisherFace, and LBPH for adaptive recognition under diverse environmental conditions.

Recognition Module
Executes real-time recognition and mask detection across dual camera inputs, ensuring continuous monitoring and verification.

Data Management Module
Aggregates recognition data, maintains records, and generates analytics reports for user tracking and system evaluation.

Face Recognition Algorithms

EigenFace, FisherFace, and LBPH Methods

- **EigenFace (PCA):** Uses Principal Component Analysis to reduce dimensionality and extract dominant facial features; efficient in controlled environments with 70–85% accuracy.
- **FisherFace (LDA):** Employs Linear Discriminant Analysis for enhanced class separation; achieves 75–90% accuracy and performs best with diverse datasets.
- **LBPH (Local Binary Patterns):** Texture-based feature extraction enabling superior robustness against lighting and angle variations; delivers 85–95% accuracy and used in production.
- **Algorithm Comparison:** LBPH excels in real-world conditions, FisherFace balances speed and precision, while EigenFace offers computational efficiency for small-scale systems.

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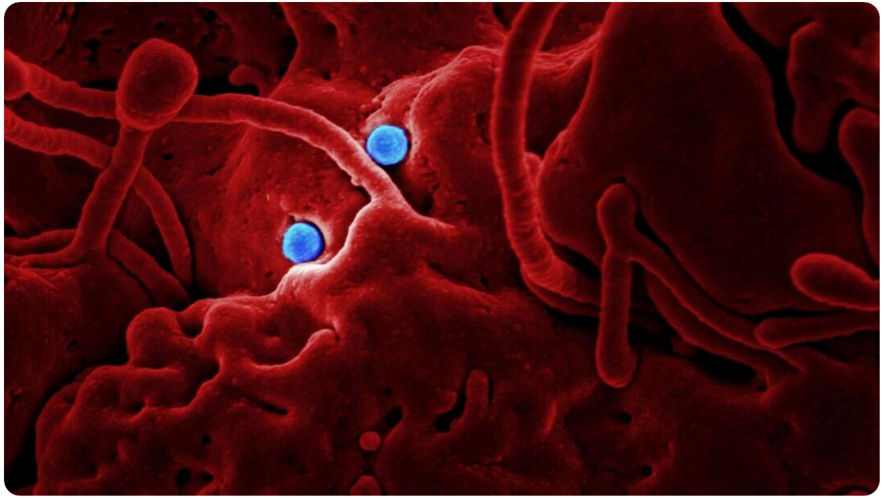
Mask Detection & Image Processing

Real-Time Detection and Preprocessing Pipeline

- **Mask Detection:** Custom-trained Haar Cascade classifier detects mask presence in real time, drawing green bounding boxes and logging compliance status.
- **Face Detection and Cropping:** System detects faces, crops the image removing outer edges to isolate features, and standardizes input for recognition models.
- **Image Enhancement:** Grayscale conversion and histogram equalization improve contrast and texture consistency across datasets.
- **Preprocessing Workflow:** All images are resized to 100×100 pixels before being passed through recognition modules ensuring uniform input dimensions.



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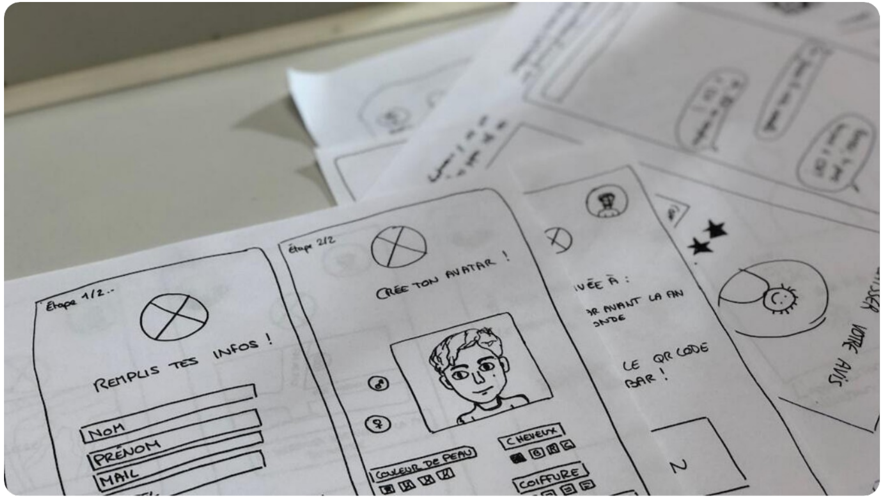
System Architecture & Workflow

Registration, Training, and Recognition Process

- **Registration Phase:** Each user registers by capturing at least 10 face images, establishing the base dataset for model training.
- **Model Training:** All three algorithms — EigenFace, FisherFace, and LBPH — are trained using preprocessed face datasets for robust recognition.
- **Real-Time Recognition:** System detects and identifies faces through video streams while performing parallel mask detection.
- **Data Management:** Captured recognition results and mask statuses are logged into CSV records for tracking and analysis.



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System Performance & Results

Accuracy, Speed, and Efficiency Metrics

- **Recognition Accuracy:** LBPH achieved 85–95% accuracy across variable lighting and pose conditions, outperforming EigenFace and FisherFace models.
- **Processing Speed:** The system processes real-time video feeds at approximately 30 frames per second, ensuring smooth user experience and minimal latency.
- **System Reliability:** Low false-positive rate with optimized memory management, enabling continuous operation with minimal resource overhead.
- **Operational Advantages:** Provides fully contactless, automated identification and mask compliance monitoring suitable for large-scale facilities.

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Technology Stack

Libraries, Algorithms, and Platform

Core Libraries
Built using OpenCV for computer vision, NumPy for numerical computation, Pandas for data management, and Tkinter for GUI design.

Recognition Algorithms
Implements Haar Cascade classifiers for face and mask detection, PCA (EigenFace), LDA (FisherFace), and LBPH for facial feature extraction.

Development Platform
Python-based modular implementation offering cross-platform compatibility and scalability for research or production use.

Integration & Extensibility
Supports future inclusion of deep learning models and database systems such as SQL or cloud storage for enterprise-grade deployments.

Conclusion & Future Enhancements

Summary, Applications, and Next Steps

System Achievements
Developed a multi-algorithm face recognition system integrating real-time mask detection with robust LBPH-based recognition.

Operational Benefits
Achieved contactless, automated identity verification with high accuracy and performance suitable for institutional deployment.

Future Improvements
Planned integration of CNN-based deep learning, database connectivity, enhanced security via encryption, and mobile app extensions.

Application Areas
Ideal for educational institutes, corporate offices, security agencies, and access control systems where safety and speed are critical.

Project Cover Page

Face Recognition & Mask Detection

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Project Title
Face Recognition and Mask Detection System

Course
Computer Vision and Artificial Intelligence